HYDRAULIC TURBINES

The hydraulic machine which converts the hydraulic energy into Mechanical Energy is called hydraulic turbine. This mechanical energy is used in running an electric generator which is directly coupled to the shaft of the hydraulic turbine. Thus, the mechanical energy is converted into Electrical energy. The electric energy which is obtained from hydraulic energy is known as hydroelectric power.

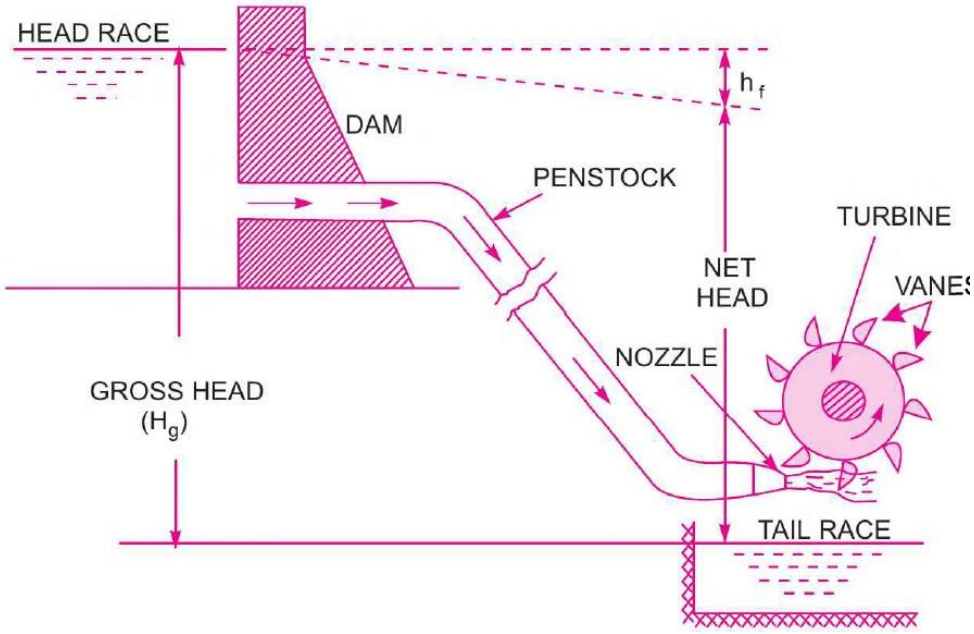


Fig: Layout of a hydroelectric power plant

Gross Head: The difference between the head race level and tail race level when no water is flowing is known as Gross head.

Net head: It is also called effective head and is defined as the head available at the inlet of the turbine.

If h_f is the head loss due to friction between penstock and water then net head on the turbine,

$$H=H_g-h_f$$

CLASSIFICATION OF HYDRAULIC TURBINES:

The hydraulic turbines are classified as follows:

1) According to the type of energy at inlet
    I. Impulse turbine

II. Reaction turbine

2) According to the direction of flow through runner
   I. Tangential flow turbine (Pelton turbine)
   II. Radial flow turbine
   III. Axial flow turbine (Kaplan turbine)
   IV. Mixed flow turbine (Francis turbine)
3) According to the head at the inlet of the turbine
   I. High head turbine
   II. Medium head turbine
   III. Low head turbine
4) According to the specific speed of the turbine
   I. Low specific speed turbine
   II. Medium specific speed turbine
   III. High specific speed turbine

1) According to the type of energy at inlet:

(i) **Impulse turbine**: Requires high head and small quantity of flow. If at the inlet of the turbine, the energy available is only kinetic energy, then the turbine is known as Impulse turbine.

(ii) **Reaction turbine**: Requires low head and high rate of flow. If at the inlet of the turbine, the energy available is kinetic energy as well as pressure energy, then the turbine is known as Reaction turbine.

2) According to the direction of flow through runner:

In tangential flow turbine of Pelton type, the water strikes the runner tangential to the path of rotation. In radial flow turbine, the water flows in radial direction through the runner. If the water flows from outwards to inwards, radially, the turbine is known as inward radial flow turbine. If the water flows from inwards to outwards, radially, the turbine is known as outward radial flow turbine. In axial flow turbine, water flows parallel to the axis of the turbine shaft. Kaplan turbine is an axial flow turbine. In mixed flow turbines, the water enters the blades radially and comes out axially, parallel to the turbine shaft. Modern Francis turbines have mixed flow runners.

3) According to the head at the inlet of the turbine:

Pelton turbine is an Impulse turbine and it is used for high head and low discharge. Francis Turbine is a Reaction turbine and it is used for medium head and medium discharge. Kaplan turbine is a Reaction turbine and it is used for low head and high discharge.

4) **According to the specific speed of the turbine:**

The specific speed of a turbine is defined as the speed of a geometrically similar turbine that would develop 1 kW under 1 m head. All geometrically similar turbines (irrespective of the sizes) will have the same specific speeds when operating under the same head.

Specific speed, Ns = $\frac{N\sqrt{P}}{H^{5/4}}$

where, N = The normal working speed,

P = Power output of the turbine, and

H = The net or effective head in metres.

Turbines with low specific speeds work under high head and low discharge conditions, while high specific speed turbines work under low head and high discharge conditions.

EFFICIENCIES OF A TURBINE:

(1) **Hydraulic Efficiency:** It is defined as the ratio of power given by water to the runner of a turbine (R.P) to the power supplied by the water (W.P) at inlet of the turbine.

(2) **Mechanical Efficiency:** It is defined as the ratio of power available at the shaft of the turbine (S.P) to the power delivered to the runner (R.P) of the turbine.

(3) **Overall Efficiency:** It is defined as the ratio of power available at the shaft of the turbine (S.P) to the power supplied by the water at the inlet(W.P)of the turbine.

(4) **Volumetric Efficiency:** It is defined as the ratio of volume of water actually striking the runner to the volume of water supplied to the runner is known as volumetric efficiency.

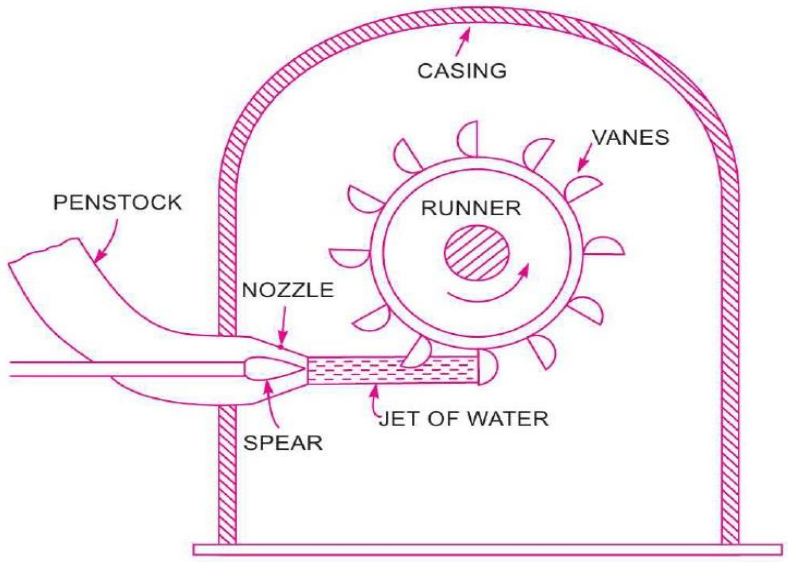
$$\mathbf{W.P=\frac{\rho g Q H}{1000}\ KW}$$

PELTON WHEEL:

Pelton turbine or Pelton wheel is a Tangential flow Impulse turbine. The water strikes the bucket along the tangent of the runner. The energy available at the inlet of the turbine is only kinetic energy. The pressure at the inlet and outlet of the turbine is atmospheric.

A Pelton wheel consists of a **runner**, at the periphery of which are mounted equally spaced **double hemispherical buckets.** Water is transferred from a high head source through penstock which is fitted with a nozzle, through which water comes out with a high velocity. A moving spear inside the nozzle controls the flow of water through the nozzle and at the same time it provides a smooth flow of water with negligible energy loss. All the available potential energy is converted into kinetic energy before striking the buckets of the runner. The pressure all over the wheel is constant and equal to atmosphere, so that energy transfer is due to purely impulse action.

The Pelton turbine is provided with a **casing,** the function of which is to prevent the splashing of water and to discharge water to the tail race. When the nozzle is completely closed by moving the spear in the forward direction, the amount of water striking the runner is reduced to zero but the runner due to inertia continues revolving for a long time. In order to bring the runner to rest in a short time, a nozzle (brake) is provided which directs the jet of water on the back of buckets; this jet of water is called **braking jet.**



Work done and Efficiency of a Pelton Wheel:

Fig. shows the velocity triangles.

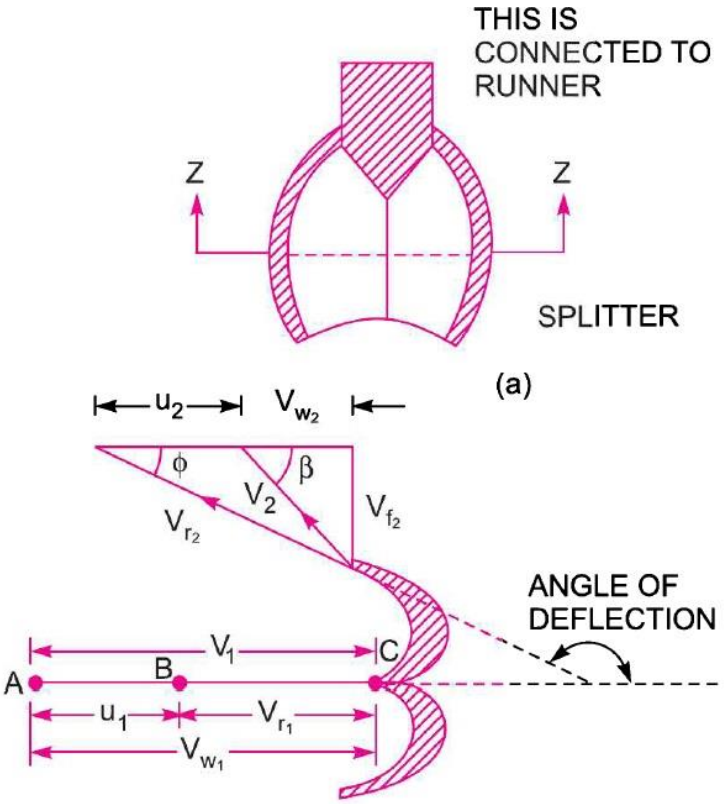
Let, N = Speed of wheel in r.p.m.,

D = Diameter of the wheel,

d = Diameter of the jet,

u = Peripheral (or circumferential) velocity of runner. It will be same at inlet and outlet of the runners. (i.e. $u = u_1 = u_2$)

$=\pi DN/60$



V_1 = Absolute velocity of water at inlet,

V_{r1} = Jet velocity relative to vane/bucket at inlet,

α = Angle between the direction of the jet and direction of motion of the vane/ bucket (also called guide angle),

θ = Angle made by the relative velocity (V_{r1}) with the direction of motion at inlet (also called vane angle at inlet),

V_{wl} and $V_{f\,1}$ = The components of the velocity of the jet V_l, in direction of motion and perpendicular to the direction of motion of the vane respectively;

V_{w1} is also known as velocity of whirl at inlet,

V_{f1} is also known as velocity of flow at inlet,

V_2 = Velocity of jet, leaving the vane or velocity of jet at outlet of the vane,

V_{r2} = Relative velocity of the jet with respect to the vane at outlet

φ = Angle made by the relative velocity V_{r2} with the direction of motion of the vane at outlet and also called vane angle at outlet,

β = Angle made by the velocity V_2 with the direction of motion of the vane at outlet,

V_{w2} and V_{f2} = Components of the velocity V_2, in the direction of motion of vane and perpendicular to the direction of motion of vane at outlet;

V_{w2} is also called the velocity of whirl at outlet, and

V_{f2} is also called the velocity of flow at outlet.

Inlet. The velocity triangle at inlet will be a straight line where

$V_{r1} = V_1 - u_1 = V_1 - u$, $V_{w1} = V_1$ (since $u_1 = u_2 = u$)

$\alpha = 0$ and $\theta = 0$

Outlet: From velocity triangle at outlet, we have

$V_{r2}=V_{r1}$

And $V_{w2}=V_{r2}\cos \phi - u_2$

Force of the jet on the turbine:

The force exerted by the jet of water in the direction of motion is given by-

$F_x=\rho a V_1 (V_{w1}+V_{w2})$

Work done by the jet:

Work done by the jet on runner per second= Fx X $u = \rho a V_1 (V_{w1}+V_{w2})$ Xu

Power given by the jet= $\{\rho a V_1 (V_{w1}+V_{w2})$ Xu$\}/1000$ KW

Work done/s per unit of water/s= $\{\rho a V_1 (V_{w1}+V_{w2})$ Xu$\}/\{1/g \times (V_{w1}+V_{w2})Xu\}$

Efficiency:

Hydraulic efficiency, η_h = Work done per second

K.E. of jet per second

$$= \frac{\rho a V1(Vw1+Vw2)X\ u}{1/2(\rho a V1)X\,(V1)2}$$

$$= \frac{2(Vw1+Vw2)X\,u}{V1^2}$$

The hydraulic efficiency will be maximum for given value of V_1 when,

$$\frac{d}{du}(\eta_h)=0$$

i.e $V_1-2u=0$

i.e $V_1=2u$

or $u=V_1/2$

The efficiency of a Pelton wheel will be maximum when the velocity of the wheel is half the velocity of jet at inlet of the turbine.

Maximum hydraulic efficiency will be $=\frac{1+\cos\phi}{2}$

Points to be remember for Pelton Wheel:

1) **The Velocity of Jet**

The velocity of the jet at inlet is given by $V_1 = C_v\sqrt{2gH}$ Where C_v = co-efficient of velocity =0.98 or 0.99. H= Net head on turbine.

2) **Velocity of Wheel**

The velocity of wheel (u) is given by $u = \phi\sqrt{2gH}$ Where, ϕ = speed ratio = 0.43 to 0.48

3) **Angle of Deflection of Jet**

The angle of deflection of jet after striking the buckets is taken as 165° if no deflection angle is given.

4) **Mean Diameter of The Wheel**

The mean diameter or the pitch diameter D of the Pelton turbine is given by

$$u = \frac{\pi DN}{60}.or.D = \frac{60u}{\pi N}$$

5) **Jet Ratio**

It is defined as the ratio of the pitch diameter (D) of the pelton turbine to the diameter of the jet (d). It is denoted by m and is given as

m = D/d

Jet ratio(m) is lies between 11 to 16 for maximum hydraulic efficiency. however, In most of the cases it is taken as 12.

6) **Number of Buckets:**

Number of buckets is given by-

Z=15+D/2d

=15+0.5m